ETAP Model Build Guide: Our Community Microgrid in ETAP 19

Oct 13, 2026 · @Res

1. Plan

We rebuild the same community network in ETAP and check that ETAP's load flow gives the same voltages and powers as our MATLAB solver. Once the two agree, ETAP can answer what our model cannot: fault currents, protection and harmonics.

Stage	ETAP study	Question it answers	Licence module
1	Balanced load flow, 3 benchmark cases	Does ETAP agree with our solver? (validates our model)	Load Flow
2	Short circuit (IEC 60909)	How big is the current in a fault at each bus? Are breakers and the 40 A trip setting sensible?	Short Circuit
3	Harmonic analysis	How much do the inverters distort the voltage? Within IEEE 519 limits?	Harmonic Analysis
4	Unbalanced load flow, single-phase	The real 230 V single-phase houses on one phase of a 3-phase feeder	Unbalanced Load Flow

This guide covers Stage 1 in full and sets up Stages 2 to 4. Work on the git branch `etap-model`; save the ETAP project in `mg_hier/docs/etap/` (an ETAP project is a folder of files; commit it when Stage 1 matches).

One honest note: the dialog and button names below are ETAP's standard ones, but I have not seen your ETAP 19 screens. If a name differs slightly, look for the nearest match and tell me what you see.

2. The per-phase equivalent: why we draw a 3-phase system

Our MATLAB model is single-phase (230 V, one live wire and neutral). ETAP's normal load flow is built for balanced 3-phase systems. The standard trick: **draw a balanced 3-phase system in which each phase is an exact copy of our single-phase system.**

What that means for the numbers:

Quantity	Our model (1-phase)	ETAP (3-phase equivalent)	Rule
Voltage	230 V line to neutral	398.4 V line to line (0.3984 kV)	ETAP is entered line to line: 230 × √3
Voltage in per unit	e.g. 0.9938	the same 0.9938	unchanged
Cable / grid impedance	0.20 + j0.04 Ω	0.20 + j0.04 Ω per phase	unchanged
Power (loads, PV, battery)	e.g. 1.5 kW	4.5 kW	× 3
Current	e.g. 6.5 A	the same 6.5 A per phase	unchanged

Use **0.3984 kV**, not 0.4 kV, everywhere. With 0.4 kV each phase would sit at 230.9 V, and every voltage would differ from ours by 0.4% before the comparison even starts.

Why this is acceptable: a balanced 3-phase system behaves like three identical single-phase systems, so the per-unit voltages, angles and currents are identical. What it cannot show is imbalance between phases; that is Stage 4.

3. Element data sheet

Every value comes from `params/hier_params.m`, with powers × 3 for the 3-phase equivalent. Use exactly these IDs in ETAP; the benchmark table in section 5 refers to them.

Buses (all 0.3984 kV, 50 Hz)

ID	What it is
`Bus-PCC`	Grid side of the main breaker
`Bus-COMM`	Community bus: battery, community solar, community load
`Bus-H1`	House 1 (prosumer)
`Bus-H2`	House 2 (consumer)

Grid and branches

ID	ETAP element	From → to	Data
`U1`	Power Grid	→ `Bus-PCC`	0.3984 kV; 3-phase short-circuit 1.100 MVA; X/R 1.5; mode Swing, 100% V, 0°
`CB-PCC`	LV Circuit Breaker	`Bus-PCC` → `Bus-COMM`	Closed in grid cases, open in the island case
`Z-H1`	Impedance	`Bus-COMM` → `Bus-H1`	R 0.20 Ω, X 0.04 Ω (enter in ohms, per phase)
`Z-H2`	Impedance	`Bus-COMM` → `Bus-H2`	R 0.30 Ω, X 0.05 Ω

Where 1.100 MVA comes from: our grid impedance is 0.08 + j0.12 Ω, so |Z| = 0.1442 Ω and X/R = 0.12/0.08 = 1.5; short-circuit MVA = kV^2 / |Z| = 0.3984^2 / 0.1442 = 1.100 MVA.

Loads (Lumped Load, set **100% motor**: ETAP then treats it as constant power, like our model)

ID	Bus	kW	Power factor	kVA
`H1-CRIT`	`Bus-H1`	0.9	0.85	1.059
`H1-FLEX`	`Bus-H1`	4.5	0.90	5.000
`H2-CRIT`	`Bus-H2`	0.9	0.85	1.059
`H2-FLEX`	`Bus-H2`	3.6	0.92	3.913
`CL-CRIT`	`Bus-COMM`	1.8	0.80	2.250

ID	Bus	kW	Power factor	kVA
`CL-PUMP`	`Bus-COMM`	4.5	0.85	5.294

Sources (Stage 1 stand-ins: Synchronous Generators at 0.3984 kV)

ID	Bus	Rating	Case A noon grid	Case B evening grid	Case C island
`G-PV-H1`	`Bus-H1`	10 kVA	Mvar control: P 6.541 kW, Q 0	out of service	out of service
`G-PV-COMM`	`Bus-COMM`	16.5 kVA	Mvar control: P 10.901 kW, Q 0	out of service	out of service
`G-BESS`	`Bus-COMM`	18 kVA	Mvar control: P 0, Q 0	Mvar control: P 9.0 kW, Q 0	**Swing**, 100% V, 0°

Why generators for now: for load flow, a generator held at a fixed P and Q behaves exactly like an inverter doing the same, and it is the simplest element to set. For short circuit and harmonics it is wrong (a generator pushes far more fault current than an inverter and makes no harmonics), so Stage 2 replaces these with ETAP's inverter elements.

PV output: 727 W per kWp at 1000 W/m² and 30 °C from our `pv_power.m`. House 1: 3 kWp × 0.727 × 3 phases = 6.541 kW. Community: 5 kWp × 0.727 × 3 = 10.901 kW.

4. Step by step in ETAP 19

4.1 New project

1. **File → New Project.** Name `URJA_Microgrid`, location `...\Commuinty_Micro_Grid\mg_hier\docs\etap\`. Unit system: **Metric**.
2. Log in as Project Editor when asked (the default user is fine on a local licence).
3. **Project → Standards:** standard **IEC**, frequency **50 Hz**.

4.2 Draw the one-line diagram (Edit mode)

You are in Edit mode when the AC Edit toolbar (bus, breaker, loads, sources) is visible on the right. Drag each element onto the page:

1. `U1` Power Grid at the top, `Bus-PCC` under it.
2. `CB-PCC` (LV Circuit Breaker) under `Bus-PCC`, then `Bus-COMM` under the breaker.
3. On `Bus-COMM`: `G-BESS`, `G-PV-COMM`, `CL-CRIT`, `CL-PUMP`.

4. Two Impedance elements hanging from `Bus-COMM`: `Z-H1` down to `Bus-H1`, `Z-H2` down to `Bus-H2`.
5. On `Bus-H1`: `G-PV-H1`, `H1-CRIT`, `H1-FLEX`. On `Bus-H2`: `H2-CRIT`, `H2-FLEX`.

Connect by dragging from one element's connection pin to the next. A connected pin turns solid; a red/unconnected element will block the study.

4.3 Enter the data

Double-click each element and fill in section 3. The tabs you need:

- **Bus:** Info → nominal kV 0.3984.
- **Power Grid:** Info → mode Swing; Rating → 0.3984 kV, 100% V, 0°; Short Circuit → 3-phase MVAsc 1.100, X/R 1.5.
- **Impedance:** Rating → units in ohms, R and X as listed, base kV 0.3984.
- **Lumped Load:** Nameplate → kV 0.3984, kW and PF as listed, **% Motor 100, % Static 0**.
- **Synchronous Generator:** Rating → 0.3984 kV, rated kVA as listed; Info/Rating → operating mode and the P, Q set-points of the case you are running.

4.4 Three cases

Use **Configuration Manager** (Configuration toolbar) to save the breaker positions: create `Grid` (CB-PCC closed) and `Island` (CB-PCC open).

The generator set-points change between cases. If your ETAP 19 does not store them per configuration, set them by hand before each run from the Sources table in section 3. Setting a PV generator to P = 0, Q = 0 is the simplest way to switch it off for cases B and C.

4.5 Run the load flow

1. Switch to **Load Flow Analysis** mode (the LF icon on the mode toolbar).
2. Open the **Study Case** editor: method **Newton-Raphson**, max iterations 99, precision 0.0001, loading category **Design**, generation category **Design**. Make one study case per benchmark case (`LF-A`, `LF-B`, `LF-C`) with the matching configuration.
3. Press **Run Load Flow**. Name the output report `LF-A` and so on.
4. On the diagram, use **Display Options** to show bus voltages in % and branch flows in kW and kvar.
5. Read the bus voltages and the grid (or battery) P and Q, and write them next to the benchmark in section 5.

If the run reports "no swing source": in the island case, `G-BESS` must be in Swing mode and `CB-PCC` open.

5. Benchmark: what ETAP should show

These answers come from our own solver (`plant/net_solve.m`), run at exactly the conditions in section 3 by `tools/etap/etap_benchmark.m` ; the numbers are also saved in `docs/etap/benchmark_cases.csv` . If ETAP lands within the tolerance below, our MATLAB power flow is independently confirmed.

Voltages are in % of 0.3984 kV (ETAP's default display); angles are relative to the swing source; powers are 3-phase totals. `Bus-PCC` reads the same as `Bus-COMM` whenever the breaker is closed.

Case	Bus-COMM	Bus-H1	Bus-H2	Source	Source P (kW)	Source Q (kvar)	ETAP result
A noon, grid	99.38% ∠ +0.31°	99.45% ∠ +0.53°	98.44% ∠ +0.46°	U1 grid	−1.18 (exporting)	+8.98	
B evening, grid	98.94% ∠ −0.06°	98.18% ∠ +0.07°	98.01% ∠ +0.09°	U1 grid	+7.30	+8.99	
C evening, island	100.00% ∠ 0°	99.25% ∠ +0.12°	99.08% ∠ +0.15°	G-BESS	+16.29	+8.98	

Pass if: every voltage within ±0.05 percentage points, every P and Q within ±0.5%. Fill in the last column as you go.

What each case tests:

- **A** checks power flowing backwards: House 1's solar exceeds its load, the community exports 1.18 kW and the house bus sits slightly above the community bus.
- **B** checks normal evening import with the battery helping: the grid supplies 7.30 kW, the battery 9 kW.
- **C** checks island mode: the battery inverter is the only source and holds the bus at 100%. In our real model the island voltage also droops with reactive power; that is switched off here so the two tools solve the same problem.

If the numbers do not match, check in this order: bus kV is 0.3984, not 0.4; loads are 100% motor (constant power), not static; the grid impedance gives 1.100 MVA with X/R 1.5; impedances are in ohms, not percent; PV generators are at Q = 0. A difference in Q only usually means a power factor typed wrongly.

6. Later stages

Stage 2: short circuit

1. Replace the three stand-in generators with ETAP's real inverter-based elements: a **PV Array + Inverter** for each solar plant, and a **Battery + Inverter** (bidirectional, "UPS/charger" style) for `G-BESS`. Give each inverter its rating (10, 16.5, 18 kVA) and a short-circuit contribution of about 1.2 × rated current, typical for inverters [typical, check the datasheet of a real product later].
2. Re-run the Stage 1 cases first: the load flow must still match section 5.
3. Switch to **Short-Circuit Analysis** mode, standard IEC 60909, 3-phase fault at each bus.

Our hand check for the grid alone (no inverter contribution), from the same ohms:

Fault at	Bolted 3-phase fault current
`Bus-COMM`	about 1,595 A
`Bus-H1`	about 713 A
`Bus-H2`	about 552 A

ETAP's IEC values will be somewhat higher (it applies a voltage factor c of about 1.05–1.10 and adds the inverters). What to learn from it: a house fault draws 550–710 A, far above our node's 40 A trip setting, so the trip will act; and in **island** mode the fault current is only what the inverter can give (about 1.2 × 26 A ≈ 31 A at 18 kVA, 3-phase), which is **below** 40 A. That is a real design question for islanded protection, and ETAP will show it with numbers.

Stage 3: harmonics

In **Harmonic Analysis** mode, give each inverter a harmonic spectrum from ETAP's library (a typical PWM inverter) and run harmonic load flow. Compare total harmonic distortion (THD) at each bus with IEEE 519 limits (at or below 1 kV: 8% total voltage THD, 5% for any single harmonic, per IEEE 519-2014). This is where our phasor model cannot follow: it only knows the 50 Hz component.

Stage 4: unbalanced, single-phase houses

With **Unbalanced Load Flow**, draw the true system: a 3-phase 0.4 kV grid, with each house as a **single-phase** load and cable on one phase (say H1 on phase A, H2 on phase B). Now the per-phase trick is gone: you will see the neutral current and the voltage imbalance that one-house-per-phase creates. This is also the right starting point for the campus model later.

Keeping it in git

When Stage 1 passes, close ETAP and commit the project folder on `etap-model` with a message such as "ETAP Stage 1: load flow matches MATLAB benchmark (A, B, C)". Add your ETAP results into the last column of the benchmark table here, so the comparison is recorded.